

MAT 425 HW # 6

1. (3 points) Consider a function $u = u(x, y)$ where the Cartesian coordinates (x, y) are related to polar coordinates (r, θ) by

$$x = r \cos \theta, \quad y = r \sin \theta.$$

- Derive an expression for u_θ in terms of partial derivatives of u with respect to x and y .
 - Using your result from part (a), compute $u_{\theta\theta}$.
2. (4 points) Let $P(x_1, x_2, \dots, x_n)$ be a homogeneous polynomial of degree d .
- Prove that for any scalar λ ,

$$P(\lambda x_1, \lambda x_2, \dots, \lambda x_n) = \lambda^d P(x_1, x_2, \dots, x_n).$$

- Provide an example of a homogeneous polynomial of degree 3 in two variables and verify the property explicitly.

Hint for part a.:

- Start by expressing $P(x_1, x_2, \dots, x_n)$ in its general form.
 - Apply the transformation $x_i \rightarrow \lambda x_i$ and simplify.
 - Use exponent properties to factor out λ^d .
 - Conclude your proof.
3. (3 points) Review and write complete definitions for the following terms from Linear Algebra:
- Linear Independence
 - Basis
 - Orthogonal
 - Orthonormal
 - The Gram-Schmidt process